University of Florida, Department of Mathematics

Analysis PhD Exam, May 1988

Answer seven out of the eight questions. Justify all work. To obtain any partial credit, all work must be presented in a neat and logical fashion.

1. Prove that $L^1(S, \mathcal{A}, \mu)$ is a complete space.

2. State and prove the Hahn–Banach theorem.

3. State and prove Fubini's theorem.

4. Prove that there exists no sequence $(c_n)_{n=1}^{\infty}$ of real numbers such that an infinite series $\sum a_n$ of real numbers converges if and only if $(c_n a_n)_{n=1}^{\infty}$ is a bounded sequence.

[Hint: Without loss of generality, assume such a sequence exists, with $c_n \neq 0$ for all n. Consider the mapping $T : \ell_\infty \to \ell_1$ given by $T((x_n)) = \frac{x_n}{c_n}$ and use the open mapping theorem.]

5. Prove that there exists no Banach space of algebraic dimension $\aleph_0$.

[Hint: show that a finite dimensional subspace is closed and then use the Baire category theorem.]

6. Let $f : \mathbb{R} \to \mathbb{R}$ be a Lebesgue measurable function. Show that there exists a Borel measurable function g such that $f = g$ a.e. (a.e. with respect to Lebesgue measure).

7. Let μ be a measure on $([0, \infty), \mathcal{B}([0, \infty)))$ defined by

$$\mu(A) = \int_A x\, dx.$$

Let $T : [0, \infty) \to [0, \infty)$ be given by $T(x) = e^x - 1$. Define a new measure π on $([0, \infty), \mathcal{B}([0, \infty)))$ by setting $\pi(A) = \mu(T^{-1}(A))$. Compute $d\pi/d\mu$.

8. Let λ be Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. If $A \in \mathcal{B}(\mathbb{R})$, define the set $-A = \{x : -x \in A\}$. Define $\mathcal{C} = \{A \in \mathcal{B}(\mathbb{R}) : A = -A\}$.

Part (a) Show $\mathcal{C}$ is a sigma-algebra.

Part (b) Define two new measures μ and ν on the measurable space $(\mathbb{R}, \mathcal{C})$ by setting

$$\mu(A) = \int_A x\, dx \qquad \text{whenever } A \in \mathcal{C},$$

$$\nu(A) = \int_{A \cap [0,\infty)} x\, dx \qquad \text{whenever } A \in \mathcal{C}.$$

Since λ may also be considered as a measure on $(\mathbb{R}, \mathcal{C})$, we can find two $\mathcal{C}$-measurable functions $d\mu/d\lambda$ and $d\nu/d\lambda$ so that

$$\mu(A) = \int_A \frac{d\mu}{d\lambda}\, dx \qquad \text{whenever } A \in \mathcal{C},$$

$$\nu(A) = \int_A \frac{d\nu}{d\lambda}\, dx \qquad \text{whenever } A \in \mathcal{C}.$$

Compute these two functions.